Physics 4B Spring 2026
Week 7 Discussion

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1. **Brainwarmer/Heat conduction**[1]. A wall of dimension 8 m × 20 m has a thickness of 15 cm. Given $\kappa = 0.8\ \frac{\text{W}}{\text{m}\cdot\text{K}}$, find the energy flowing across in a day if the temperature difference is 15°C.

   Solution: The rate of heat conduction (from hot to cold) is

   $$\frac{dQ}{dt} = \kappa \frac{A\Delta T}{L} \tag{1}$$

   The total energy flow over a day ($\Delta t = 86400$ s) is

   $$\begin{aligned} Q = \frac{dQ}{dt}\Delta t &= \left(0.8\ \frac{\text{W}}{\text{m}\cdot\text{K}}\right)\cdot\left(160\ \text{m}^2\right)\cdot(15\ \text{K})\cdot\left(\frac{1}{0.15\ \text{m}}\right)\cdot(86400\ \text{s}) \\ &= 1.1\times 10^9\ \text{J}. \end{aligned} \tag{2}$$

2. **Air bubble**[2]. A spherical air bubble of radius 2 cm is released 30 m below the surface of a pond at 280 K. What is its volume when it reaches the surface, which is at 300 K, assuming it is in thermal equilibrium the whole time? Use $P_{\text{atm}} = 101325$ Pa. *Ignore the size of the bubble compared to other dimensions like 30 m.*

   Solution: We are going to treat the air in the bubble as an ideal gas. The air starts in a state with pressure

   $$P_1 = P_{\text{atm}} + \rho_w g h = 101325\ \text{Pa} + \left(1000\ \frac{\text{kg}}{\text{m}^3}\right)\cdot(9.8\ \text{m/s}^2)\cdot(30\ \text{m}) = 395325\ \text{Pa}, \tag{3}$$

   volume $V_1 = \frac{4}{3}\pi r^3$ and temperature $T_1 = 280$ K. The air ends in a state with pressure $P_2 = P_{\text{atm}}$, unknown volume V_2 and temperature $T_2 = 300$ K. Since the ideal gas law $PV = Nk_BT$ holds in both the initial and final states, we have

   $$\frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2} \tag{4}$$

   Therefore,

   $$V_2 = \frac{P_1V_1T_2}{P_2T_1} = 1.4\times 10^{-4}\ \text{m}^3 \tag{5}$$

[1] Shankar 11.9

[2] Shankar 11.14

3. **Kinetic theory**[3]. In class we learned that the Maxwell-Boltzmann velocity distribution is

$$f(\vec{v}) \propto \exp\left(-\frac{mv^2}{2k_BT}\right) \tag{6}$$

where $v^2 = v_x^2 + v_y^2 + v_z^2$. This comes from assuming the x, y, z components of the velocity are independent and that there's no preferred direction of motion for the gas particles.

(a) Find the normalization constant of the distribution such that $\int d^3\vec{v}\, f(\vec{v}) = 1$. *Hint*: $\int_{-\infty}^{\infty} dx\; e^{-ax^2} = \sqrt{\pi/a}$.

Solution: Let the normalization constant be C and $a \equiv m/2k_BT$, so

$$\begin{aligned} 1 &= \int d^3\vec{v}\; f(\vec{v}) = C\int_{-\infty}^{\infty} dv_x \int_{-\infty}^{\infty} dv_y \int_{-\infty}^{\infty} dv_z \exp\left(-a(v_x^2 + v_y^2 + v_z^2)\right) \\ &= C\left(\frac{\pi}{a}\right)^{3/2} = C\left(\frac{2\pi k_BT}{m}\right)^{3/2} \end{aligned} \tag{7}$$

Therefore, the normalization constant is

$$C = \left(\frac{m}{2\pi k_BT}\right)^{3/2} \tag{8}$$

(b) Show that the Maxwell-Boltzmann speed distribution is given by

$$\tilde{f}(v) \propto v^2 \exp\left(-\frac{mv^2}{2k_BT}\right) \tag{9}$$

Solution: The fraction of gas particles that have v_x in dv_x, v_y in dv_y and v_z in dv_z is by definition

$$N = f(\vec{v})\; d^3\vec{v} = f(\vec{v})\; dv_x\; dv_y\; dv_z \tag{10}$$

By a similar definition, the fraction of gas particles that have speeds v in dv is $\tilde{f}(v)\; dv$.

If we look at the velocity space (v_x, v_y, v_z), velocities that have the same speed form a spherical shell with radius $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ centered at the origin, so the volume of the shell is $d^3\vec{v} = 4\pi v^2 dv$, and

$$N = f(\vec{v})\; 4\pi v^2 dv = \tilde{f}(v)\; dv \tag{11}$$

[3]Adapted from additional kinetic theory homework problems

Therefore, the speed distribution is

$$\tilde{f}(v) = 4\pi v^2 f(\vec{v}) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) \tag{12}$$

For simplicity, we will drop the tilde and write $\tilde{f}(v)$ simply as $f(v)$ from now on.

(c) Using the speed distribution you just found, find the most probable speed $v_{\rm mp}$, the average speed $\langle v\rangle$, and the root-mean-square speed $v_{\rm rms} = \sqrt{\langle v^2\rangle}$. *Hint*: $\int_0^\infty dx\ x^3 e^{-ax^2} = 1/2a^2$; $\int_0^\infty dx\ x^4 e^{-ax^2} = \frac{3}{8}\sqrt{\pi/a^5}$.

Solution: The most probable speed corresponds to the maximum value of the distribution. Again let $a \equiv m/2k_B T$,

$$0 = \frac{df(v)}{dv} \propto 2e^{-av^2} v(1 - av^2) \tag{13}$$

The reasonable solution is

$$v_{\rm mp} = a^{-1/2} = \sqrt{\frac{2k_B T}{m}} \tag{14}$$

The average speed can be calculated using

$$\begin{aligned}\langle v\rangle &= \int_0^\infty dv\ v f(v) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi \int_0^\infty dv\ v^3 \exp\left(-\frac{mv^2}{2k_B T}\right) \\ &= \left(\frac{a}{\pi}\right)^{3/2} 4\pi \int_0^\infty dv\ v^3 e^{-av^2} = \left(\frac{a}{\pi}\right)^{3/2} 2\pi a^{-2} = 2\pi^{-1/2} a^{-1/2} \\ &= \sqrt{\frac{8k_B T}{\pi m}}\end{aligned} \tag{15}$$

Similarly, the rms speed is

$$\begin{aligned}v_{\rm rms} &= \sqrt{\int_0^\infty dv\ v^2 f(v)} = \sqrt{\left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi \int_0^\infty dv\ v^4 \exp\left(-\frac{mv^2}{2k_B T}\right)} \\ &= \sqrt{\left(\frac{a}{\pi}\right)^{3/2} 4\pi \int_0^\infty dv\ v^4 e^{-av^2}} = \sqrt{\left(\frac{a}{\pi}\right)^{3/2} \frac{3}{2}\pi^{3/2} a^{-5/2}} = \sqrt{\frac{3}{2}a^{-1}} \\ &= \sqrt{\frac{3k_B T}{m}}\end{aligned} \tag{16}$$

(d) Compute the average value of $|v_x|$ for an ideal gas. How does it compare to the average speed $\langle v\rangle$? *Hint*: $\int_0^\infty dx\ xe^{-ax^2} = 1/2a$.

Solution: This requires us to look at the distribution for v_x only, which is simply $f(\vec{v})^{1/3}$:

$$f(v_x) = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left(-\frac{mv_x^2}{2k_B T}\right) \tag{17}$$

Following the same procedure as in the previous part, we have

$$\begin{aligned}\langle |v_x| \rangle &= \int_{-\infty}^{\infty} dv_x \ |v_x| f(v_x) = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \int_{-\infty}^{\infty} dv_x \ |v_x| \exp\left(-\frac{mv_x^2}{2k_B T}\right) \\ &= \left(\frac{a}{\pi}\right)^{1/2} 2\int_0^{\infty} dv_x \ v_x e^{-av_x^2} = \left(\frac{a}{\pi}\right)^{1/2} a^{-1} = \sqrt{\frac{2k_B T}{\pi m}} = \frac{1}{2}\bar{v}\end{aligned} \tag{18}$$

(e) In a gas of helium atoms ($m_{\text{He}} = 6.646 \times 10^{-27}$ kg) at 300 K, there are 10,000 atoms that have velocity in the x direction between -1 m/s and 1 m/s. Approximately how many atoms have velocity in the y direction between 100 m/s and 101 m/s?

Solution: This means

$$10000 = N \cdot \int_{-1}^{1} dv_x f(v_x) = N \left(\frac{m}{2\pi k_B T}\right)^{1/2} \int_{-1}^{1} dv_x \exp\left(-\frac{mv_x^2}{2k_B T}\right) \tag{19}$$

We can evaluate the integral to determine N, the total number of helium atoms, using the mass and temperature given to us. The number of atoms that have $v_y \in [100, 101]$ m/s would be

$$\begin{aligned}N \cdot \int_{100}^{101} dv_y f(v_y) &= N \left(\frac{m}{2\pi k_B T}\right)^{1/2} \int_{100}^{101} dv_y \exp\left(-\frac{mv_y^2}{2k_B T}\right) \\ &= 10000 \cdot \frac{\int_{100}^{101} dv_y \exp\left(-\frac{mv_y^2}{2k_B T}\right)}{\int_{-1}^{1} dv_x \exp\left(-\frac{mv_x^2}{2k_B T}\right)}\end{aligned} \tag{20}$$

Now, since the intervals we are considering ($v_x \in [-1, 1]$ m/s and $v_y \in [100, 101]$ m/s) are narrow, we can approximate the above integrals by taking the integrand to be roughly constant over the interval:

$$\begin{aligned}10000 \cdot \frac{\int_{100}^{101} dv_y \exp\left(-\frac{mv_y^2}{2k_B T}\right)}{\int_{-1}^{1} dv_x \exp\left(-\frac{mv_x^2}{2k_B T}\right)} &\approx 10000 \cdot \frac{\Delta v_y \cdot \exp\left(-\frac{m\bar{v}_y^2}{2k_B T}\right)}{\Delta v_x \cdot \exp\left(-\frac{m\bar{v}_x^2}{2k_B T}\right)} \\ &= 10000 \cdot \frac{1}{2} \exp\left(\frac{m(\bar{v}_x^2 - \bar{v}_y^2)}{2k_B T}\right) \\ &\approx 5000\end{aligned} \tag{21}$$

(f) Assume there is a very small hole in the wall of a container that contains an ideal gas. Show that the mean kinetic energy of the molecules that escape is larger than $\frac{3}{2}k_BT$. What is it?

Solution: Inside the container, the average kinetic energy per molecule is $\frac{3}{2}k_BT$, which is equally split between the 3 directions:

$$\frac{1}{2}m\langle v_x^2\rangle_{\text{inside}} = \frac{1}{2}m\langle v_y^2\rangle_{\text{inside}} = \frac{1}{2}m\langle v_z^2\rangle_{\text{inside}} = \frac{1}{2}k_BT \tag{22}$$

However, the molecules that escape from the hole are biased and are not a random sample - particles that have large velocity component perpendicular to the wall with the hole are more likely to escape, so (22) doesn't apply to them. For these particles, we need to make different assumptions to compute the rms velocities $\langle v_i^2\rangle$ $(i = x, y, z)$.

Let the direction perpendicular to the wall with the hole be z. It would be reasonable to argue that faster velocities in the x, y directions don't make the particles more likely to escape the hole, so the average kinetic energy in these directions for the *escaping particles* should remain the same as those inside the container:

$$\frac{1}{2}m\langle v_x^2\rangle_{\text{esc}} = \frac{1}{2}m\langle v_y^2\rangle_{\text{esc}} = \frac{1}{2}k_BT \tag{23}$$

For the z direction, the probability of a particle hitting the wall is directly proportional to v_z, since a particle with twice the value of v_z would hit the wall twice as often. Also, particles that escape must have $v_z > 0$. Therefore, the distribution for the z-component of the velocity for the *escaping particles* should take the form

$$f_{\text{esc}}(v_z) \propto v_z \exp\left(-\frac{mv_z^2}{2k_BT}\right) \tag{24}$$

with $v_z \in (0, \infty)$. The normalization factor is determined by the condition $1 = \int_0^\infty dv_z\ f_{\text{esc}}(v_z)$. Using this, we can compute the average kinetic energy in the z direction for the escaped particles (again we define $a \equiv m/2k_BT$):

$$\begin{aligned}\frac{1}{2}m\langle v_z^2\rangle_{\text{esc}} &= \frac{1}{2}m \cdot \frac{\int_0^\infty dv_z\ v_z^2 f_{\text{esc}}(v_z)}{\int_0^\infty dv_z\ f_{\text{esc}}(v_z)} = \frac{1}{2}m \cdot \frac{\int_0^\infty dv_z\ v_z^3 e^{-av_z^2}}{\int_0^\infty dv_z\ v_z e^{-av_z^2}} \\ &= \frac{1}{2}m \cdot \frac{\frac{1}{2}a^{-2}}{\frac{1}{2}a^{-1}} = \frac{1}{2}ma^{-1} = k_BT\end{aligned} \tag{25}$$

Therefore, the mean kinetic energy of the escaped particles is

$$\frac{1}{2}m\langle v^2\rangle_{\text{esc}} = \frac{1}{2}m\left(\langle v_x^2\rangle_{\text{esc}} + \langle v_y^2\rangle_{\text{esc}} + \langle v_z^2\rangle_{\text{esc}}\right) = 2k_BT > \frac{3}{2}k_BT \tag{26}$$

4. **Ideal nitrogen gas**[4]. One mole of ideal nitrogen gas is at 2 atmospheres and occupies a volume of 10 m^3. Find (i) the temperature T in Kelvin, (ii) the internal energy U (assumed to be just kinetic energy) per molecule in joules, and (iii) the typical velocity of the gas molecules, which have a mass 4.65×10^{-26} kg.

Solution:

(i) One mole corresponds to $N = N_A = 6.02 \times 10^{23}$. We use the ideal gas law to find the temperature

$$T = \frac{PV}{Nk_B} = \frac{(2 \times 101325 \text{ Pa}) \cdot (10 \text{ m}^3)}{(6.02 \times 10^{23}) \cdot (1.38 \times 10^{-23} \text{ J/K})} \approx 2.4 \times 10^5 \text{ K} \tag{27}$$

(ii) If we assume the ideal nitrogen gas consists of monatomic molecules, then there are 3 degrees of freedom per molecule and the internal energy is

$$U = \frac{3}{2} k_B T = 5.05 \times 10^{-18} \text{ J}. \tag{28}$$

(iii) Assume by typical velocity we mean the rms velocity, then

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3 \cdot (1.38 \times 10^{-23} \text{ J/K})(2.4 \times 10^5 \text{ K})}{4.65 \times 10^{-26} \text{ kg}}} \approx 14737 \text{ m/s} \tag{29}$$

[4]Shankar 11.16